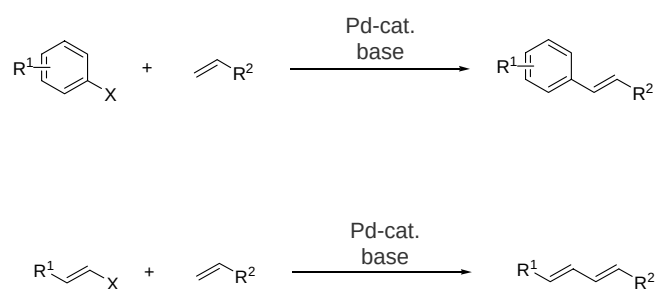


General reaction scheme



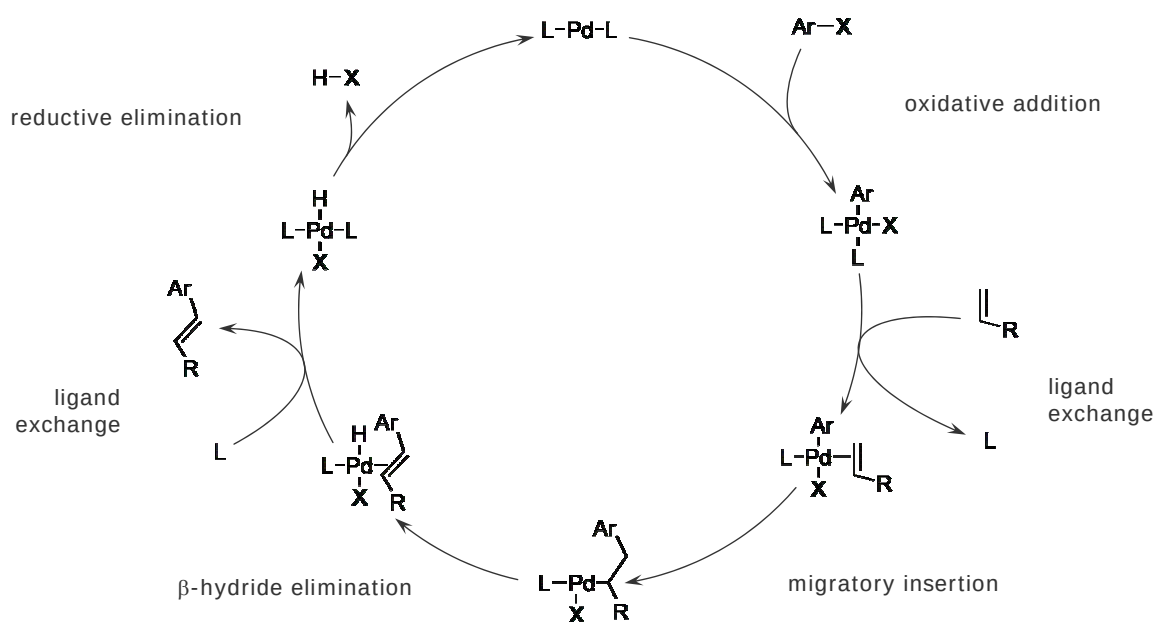
Initial publications

Mizoroki *Bull Chem Soc Jpn* **1971** 44 581

Heck *JOC* **1972** 37 2320

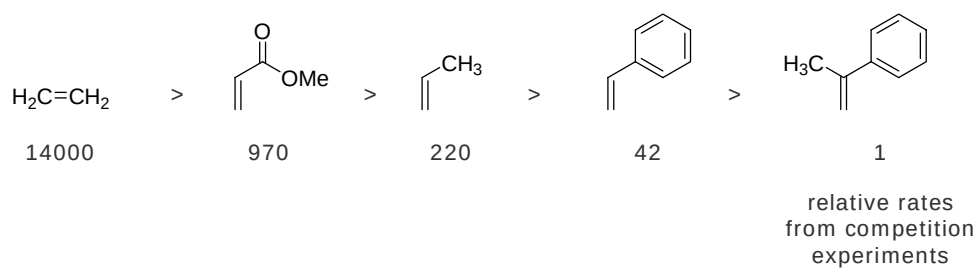
Heck *JACS* **1974** 96 1133

Heck reaction: Mechanism

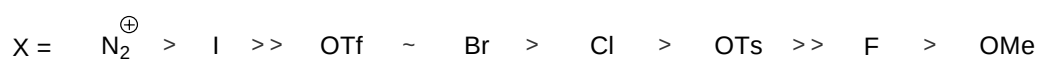


Heck Reaction: Reactivity

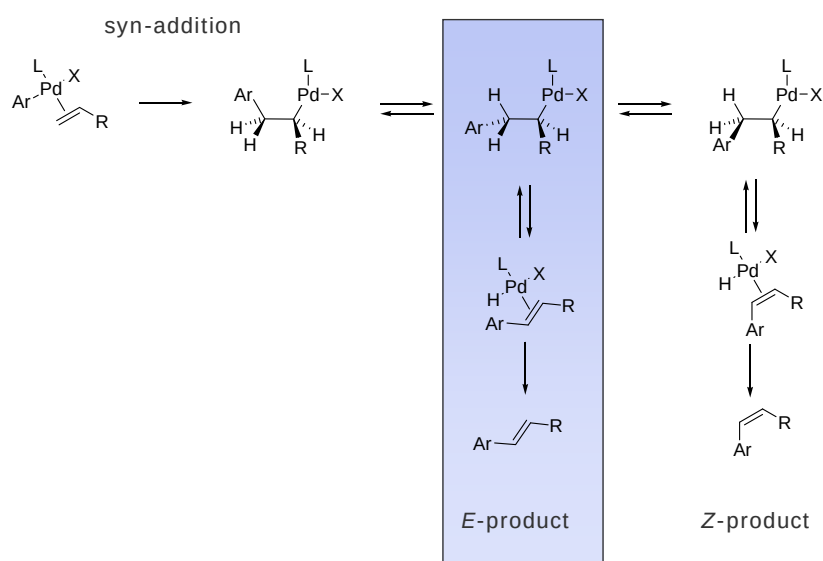
Olefin:



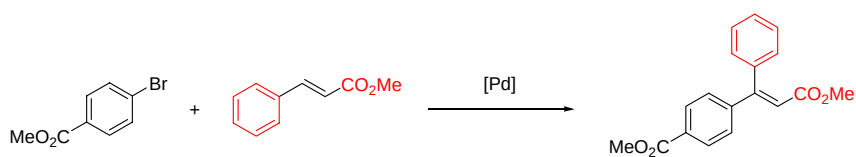
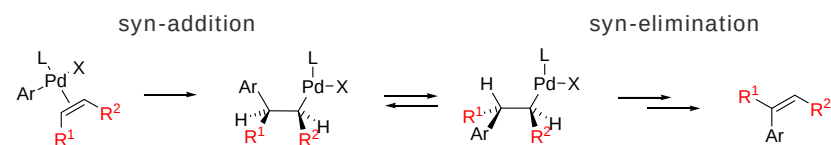
Ar-X:



Heck Reaction: Stereochemistry

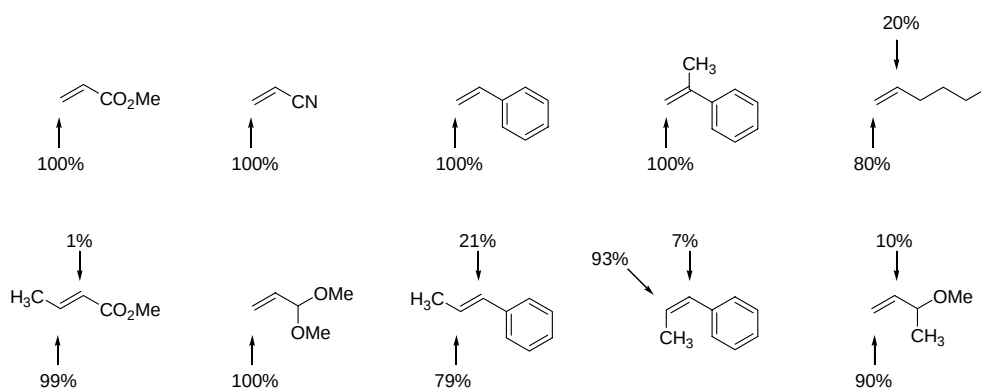
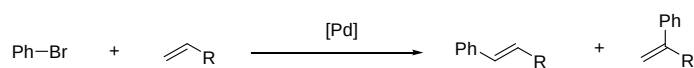


Heck Reaction: Stereochemistry



make trisubstituted olefins
stereoselectively
but: beware of base-induced
isomerization

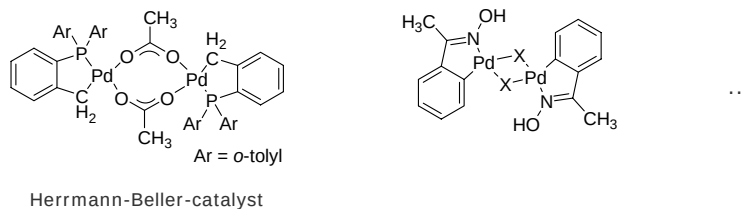
Heck Reaction: Regiochemistry



Heck Reaction: Conditions

palladium sources: $\text{Pd}(\text{OAc})_2$, $\text{Pd}_2(\text{dba})_3$

ligands: PPh_3 , *o*-Tol₃P, *t*Bu₃P, NHC...



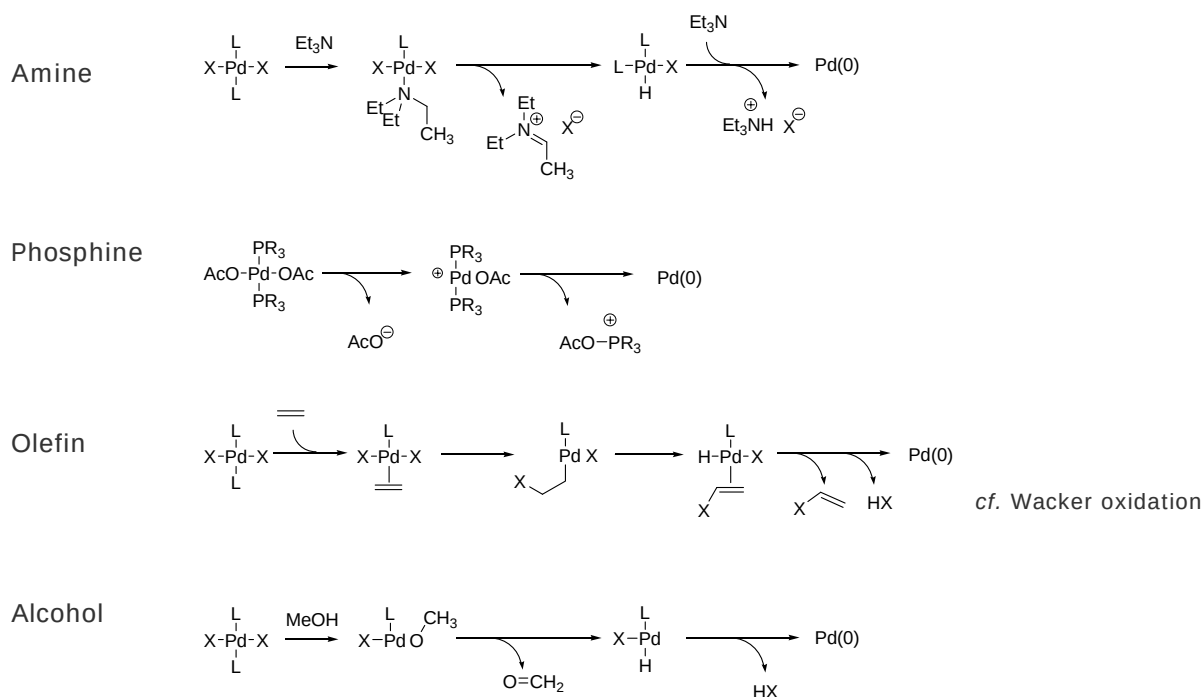
catalyst loading: 0.1 - 5 mol%, but examples with TON ~ 1 000 000

bases: Et_3N , Cy_2NEt , NaOAc , Na_2CO_3 , NaHCO_3 ,...

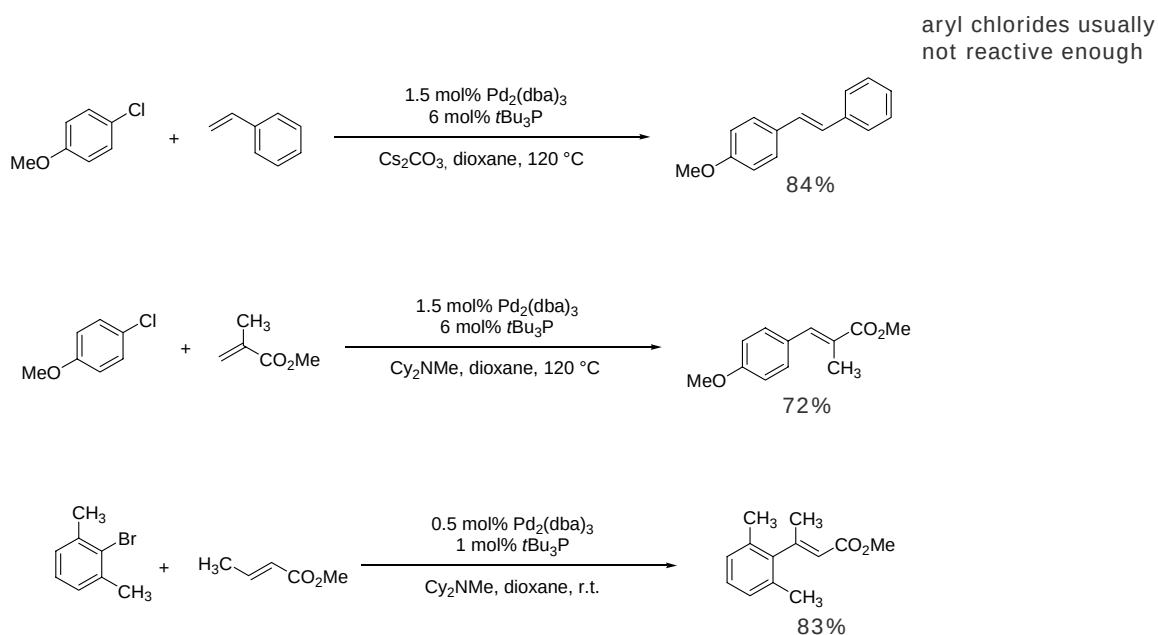
solvents: DMF, NMP, DMA, dioxane, toluene, xylene, CH_3CN ,...

temperature: usually > 100 °C

Reduction of Palladium

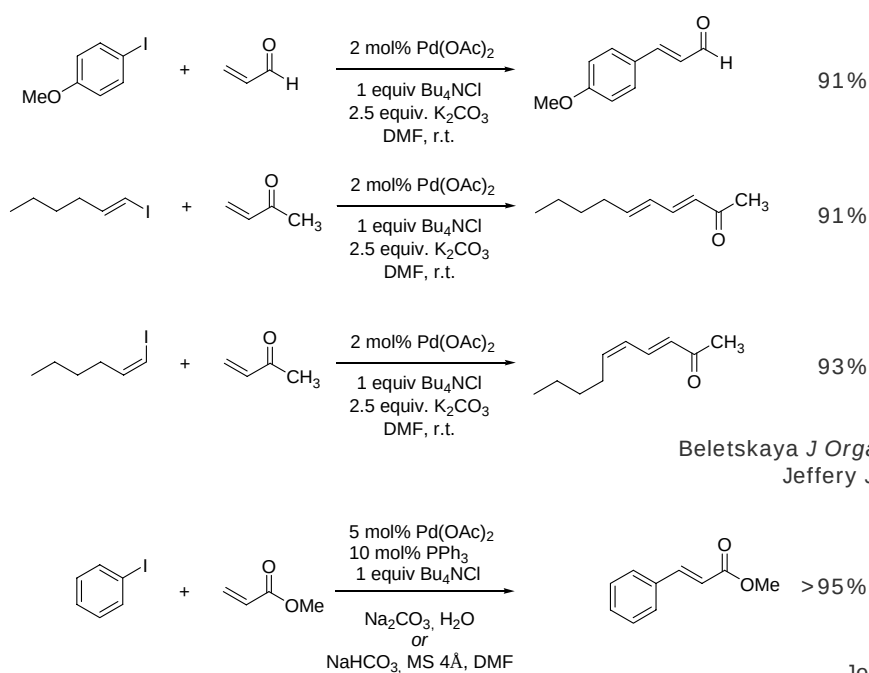


Heck Reaction: Aryl Chlorides



Fu JOC **1999** 64 10
JACS **2001** 123 6989

Heck Reaction: Jefferey Conditions



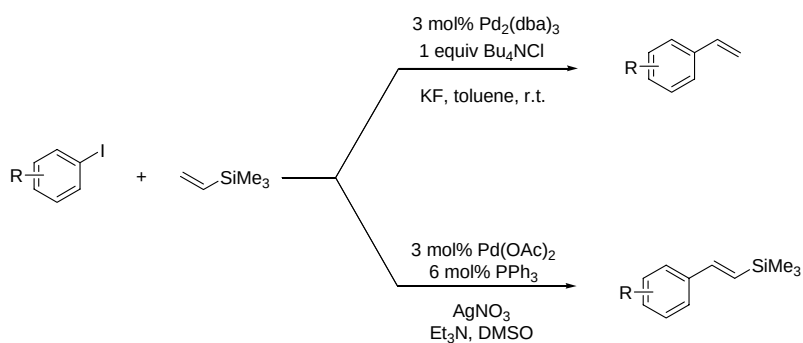
Bu₄NCl:

- increase reactivity
- lower temperature
- use sensitive alkenes

Beletskaya J Organomet Chem **1983** 250 551
Jeffery JCS Chem Comm **1984** 1287
THL **1985** 26 2667

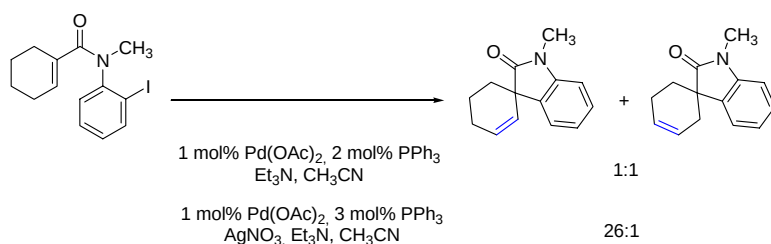
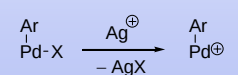
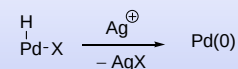
Jeffery THL **1994** 35 3051, 4103
TH **1996** 52 10113

Heck Reaction: Silver Effect



Silver salts:

- prevent double bond isomerization
- increase rate of reaction

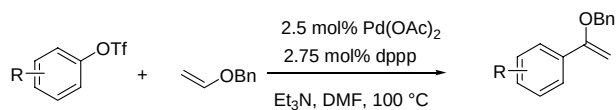


Hallberg JOC **1985** 50 3896
 JOC **1986** 51 5286
 Overman JOC **1987** 52 4130

C.C. Tzschucke

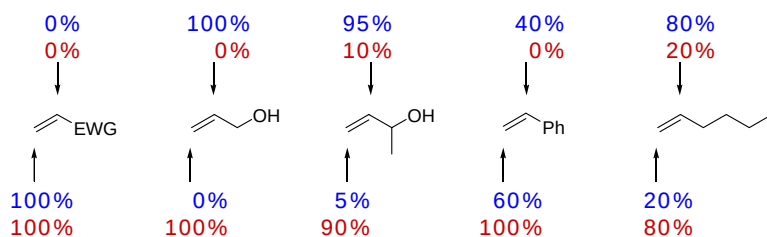
38

Heck Reaction: Triflates



- inversed regiochemistry
- cationic mechanism
- chelating ligands effective
- similar to use of Ag salts

comparison of regiochemistry
 for reaction of **aryl triflates** and **aryl halides**:

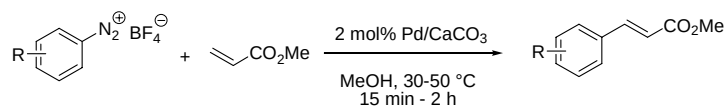


Cabri Acc Chem Res **1995** 28 2
 Hallberg TH **1994** 50 285
 JOC **2000** 65 7235

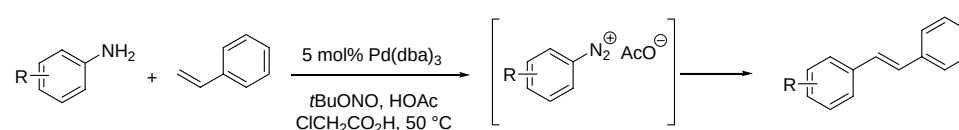
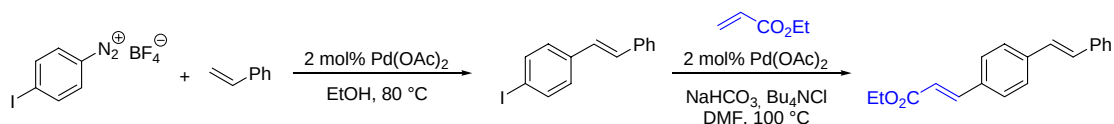
C.C. Tzschucke

40

Heck Reaction: Diazonium Salts

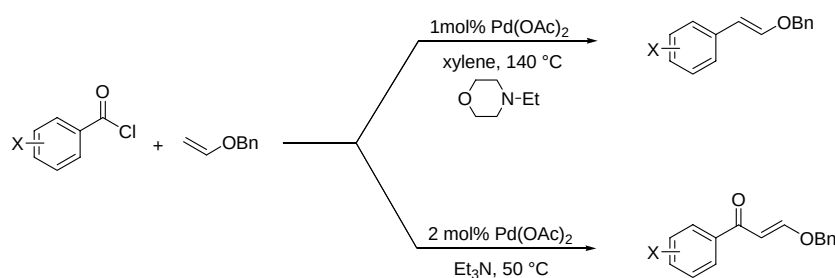


- fast reactions
- high catalyst loadings
- no base necessary
- no phosphine ligands



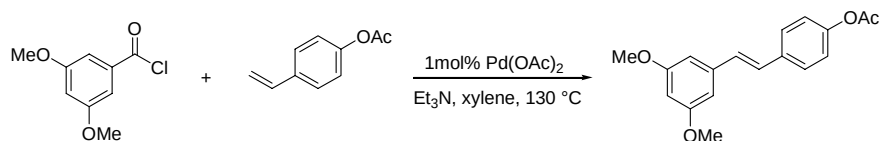
Genet *THL* **1999** 40 4815
 Sengupta *THL* **1998** 39 715
 Matsuda *JOC* **1981** 46 4885

Heck Reaction: Acid Chlorides

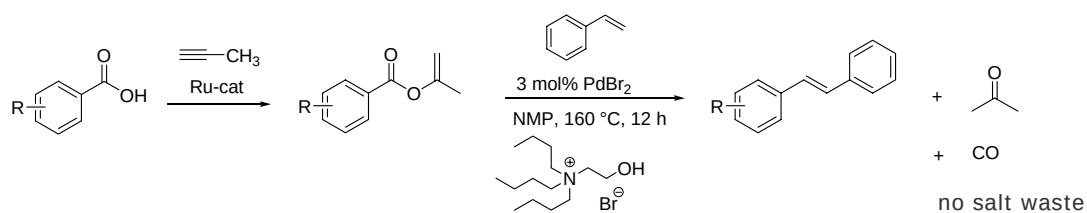


- electron withdrawing substituents favor decarbonylation
- phosphines inhibit reaction

Blaser *JOMC* **1982** 233 267
 Hallberg *JOC* **1988** 53 235
JOC **1988** 53 4257

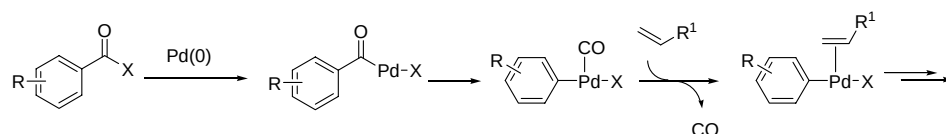


Decarbonylative Heck Reaction



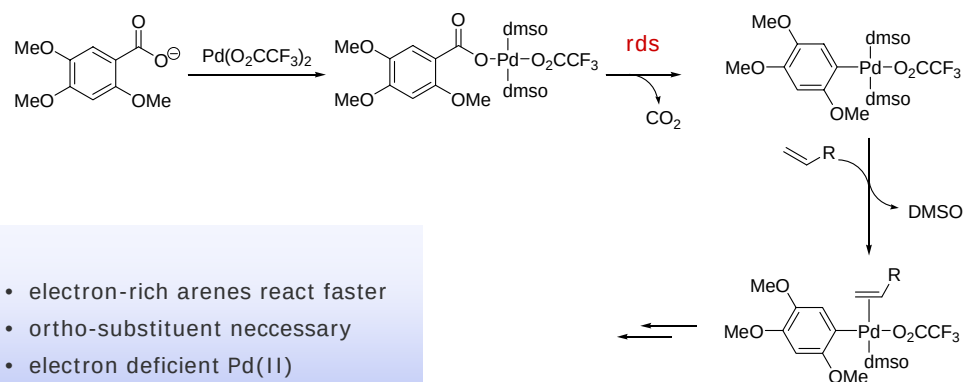
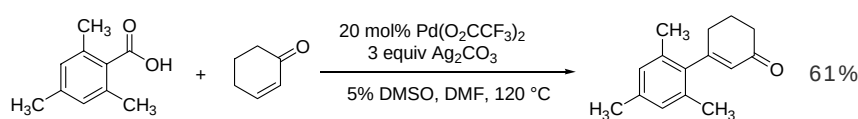
similar: activated esters, anhydrides

mechanism:



de Vries *ACIE* **1998** 37 662
Goossen *Synlett* **2002** 1721
Angew Chem **2004** 116 1115

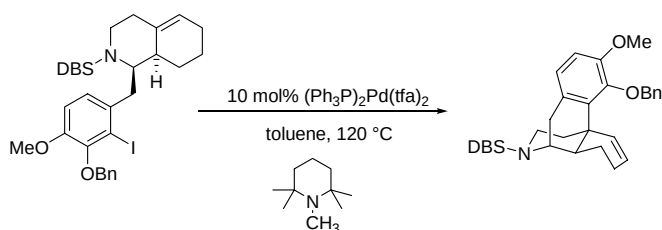
Decarboxylative Heck Reaction



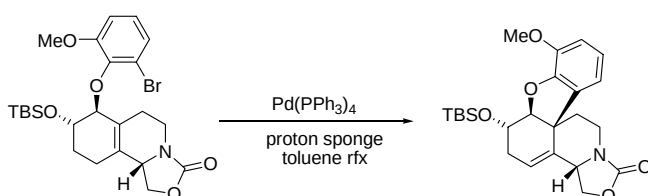
- electron-rich arenes react faster
- ortho-substituent necessary
- electron deficient Pd(II)
- electron-rich olefins react faster

Myers *JACS* **2002** 124 11250
JACS **2005** 127 10323

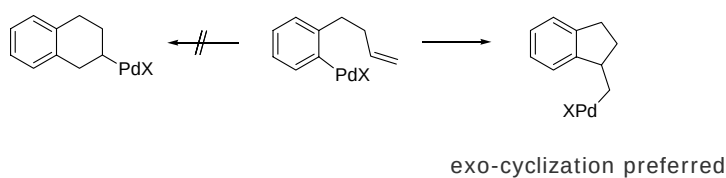
Heck Reaction: Intramolecular



Overman JACS **1993** 115 11028



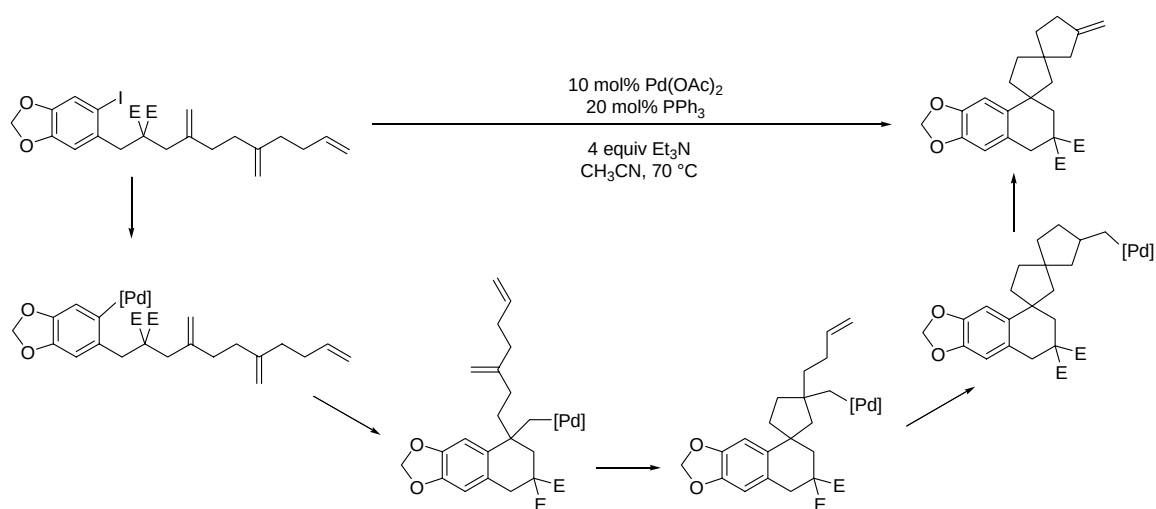
Hudlicky OL **1999** 1 2085



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47

Heck Reaction: Zipper



Overman JACS **1988** 110 2328

C.C. Tzschucke

48